

SUNSPOKE · REQUIREMENTS

SunSpoke requirements

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DOCUMENT CONTROL

Document ID	SSP-REQ-001	Version	0.2
Status	Draft	Date	2026-09-24
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/sunspoke		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-24	Amish Chadha	First measurable requirements for TRL 2
0.1	2026-09-24	Amish Chadha	Initial scaffold

SunSpoke requirements

These are first-pass requirements for the concept. Targets are proposals for review, not yet validated with users, and will be checked by calculation at TRL 3 and revised after co-design sessions (see SSP-PRB-001).

The **design load case** used throughout is a 75 kg rider, 25 kg of cargo on the rear carrier, a 22 kg roadster and about 6.5 kg of kit and pack: about 130 kg in total, on a dry dirt road.

ID	Requirement	Target	Verification (TRL 3 or later)
R1	Pedal-assist power within pedelec limits	250 W rated continuous motor output; assist only while pedaling; assist cuts off at 25 km/h or lower	Motor and controller datasheets; controller settings
R2	Range per charge in the design load case	30 km or more on a dirt road with the rider pedaling at about 70 W	Energy calculation; later field ride with a logger
R3	Climb a loaded hill	8 % grade for 500 m at 8 km/h or more in the design load case, rider at 80 W, without motor over-temperature cutout at 35 °C ambient	Force and thermal calculation; later hill test
R4	Fit to the donor bike without fabrication	No welding, drilling or cutting of the frame or fork; fitted in 90 min or less by a trained local mechanic with basic tools (spanners to 18 mm, hex keys, spoke key, screwdriver, file)	Fitting sequence review; later timed fitting trials
R5	Fit the common roadster	28 in (ETRTO 635) front wheel, 100 mm front dropout spacing, round down tube 28 to 32 mm, rod or cable brakes	Survey of donor bikes with the partner; model check
R6	Keep the bike rideable and liftable	Added mass 7 kg or less including the pack; bike still stands on its kickstand and can be pushed with the system off	Mass estimate; later weighing
R7	Survive dust, rain and heat	Controller, host adapter and connectors IP65; motor IP54 or better; ride through heavy rain and 150 mm of standing water; operate at 0 to 45 °C ambient	Datasheets and design review; later spray and dust tests
R8	Charge from one 100 W solar panel	Energy for 20 km or more of loaded assist per day at 4.5 peak sun hours; full recharge (10 to 100 %) of the pack in 2 days or less	Solar yield calculation
R9	Repairable by a local mechanic	Every electrical joint pluggable with keyed connectors; any single kit part swapped in 20 min or less at the roadside; basic faults diagnosed with a multimeter and a printed fault chart; 70 % or more of kit cost in generic parts available in regional towns	Design review; parts availability survey with the partner
R10	Safe assist control	Brake cut-off on both brake levers; assist stops within 0.5 s of braking or of pedaling stopping; pack output fused; torque arms on both fork dropouts	Design review; later bench test
R11	Stop safely with the added mass and speed	Design load case stops from 20 km/h in 9 m or less on a dry dirt road	Braking calculation; later field test with the donor brakes
R12	Affordable	Kit including the solar charging set, pack excluded, \$250 or less in parts	Priced BOM (bom/ bom.csv)

■ Assumptions

- Rolling resistance coefficient about 0.02 on a dry dirt road; drag area about 0.6 m² for an upright rider with cargo.
- The rider supplies about 70 W while cruising and 80 W on a climb.

- 4.5 peak sun hours per day is a mid-range value for most of sub-Saharan Africa outside the rainy season; 4 hours is used as the low case.
- Pack capacity follows the SwapCell interface v0.2 (about 468 Wh). A separate 36 V pack option is assessed in SSP-PRC-001.
- A 25 km/h assist limit follows EN 15194; a lower limit may be proposed after field input on braking.